

Multiphysiological Shallow Neural Network-Based Mental Stress Detection System for Wearable Environment

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Abstract—Health problems related to stress are increasing globally and significantly affect the mental health and quality of life of human beings. Continuous suffering from stress may lead to serious psychological and physical health problems. But still, no effective and reliable stress detection methods are available. In this paper, a novel wearable device is presented to measure electroencephalogram (EEG) and electrocardiogram (ECG) simultaneously in a non-invasive approach. This system includes an analog front end (AFE) integrated with a machine learning-based digital backend (DBE) processor for mental stress prediction using only 3 electrodes. A PCB prototype is developed using the commercial off-the-shelf components. The developed prototype shows excellent noise performance of $0.1\mu V_{rms}$ and predicts the mental stress with a classification accuracy of 92.7%. The proposed system is lightweight and easily wearable (behind the ear). The data is acquired from 25 participants for different stress scenarios including the Arithmetic Test and Stroop Color Word Test. Different EEG and ECG based features combinations are used for the classification of stress conditions using a shallow neural network (SNN) classifier.

Index Terms—Electroencephalogram (EEG), Electrocardiogram (ECG), Heart Rate Variability (HRV), Analog Front End (AFE), Stress Estimation, Mental Arithmetic Test (MAT), Stroop Color Word Test (SCWT), Shallow Neural Network (SNN) Classifier.

I. INTRODUCTION

Stress is a reaction of the human body to a challenging situation disturbing the mental equilibrium. It is induced by mental, physical, or emotional factors [1] and affects the health and life of a person [2]. Stress is broadly classified into acute and chronic stress. Acute stress is the immediate body response to the stressor and is usually for a short duration. The majority of people experience it in daily life activities including arithmetic challenges, examinations, meeting new people, and decision making [3]. Chronic stress is caused due to prolonged events in the recent past and usually arises from socioeconomic conditions, unsatisfying careers, wars, economic crises, and poverty [4].

The persistence of stress for a longer duration can not only cause various chronic diseases (high blood pressure, heart attack etc), but can also lead to early death [5], [6]. Medical research estimates that $\approx 90\%$ of illnesses and diseases relate to mental stress [7]. Therefore, it is quite important to effectively manage and control human stress after timely prediction to avoid serious health issues [8] - [10].

Stress detection is a challenging task in the absence of standard stress measurement bio-markers. The stress assessment methods include blood pressure, vagal tone and saliva tests. However, there is a dire need of a real-time, continuous, wearable and non-invasive stress prediction device [11].

This work targets a wearable, non-invasive and real-time stress prediction system using Electroencephalogram (EEG) and Electrocardiogram (ECG) signals. The proposed system on chip (SoC) device includes a novel Analog Front End (AFE) for data acquisition and a machine learning based digital backend processor for real-time stress prediction.

Fig. 1 shows the electrodes placement of the person for stress detection. Two EEG electrodes (TP9, TP10) are placed at the bottom of mastoids on both left and right sides of the head and the reference electrode was placed at the center of the forehead.

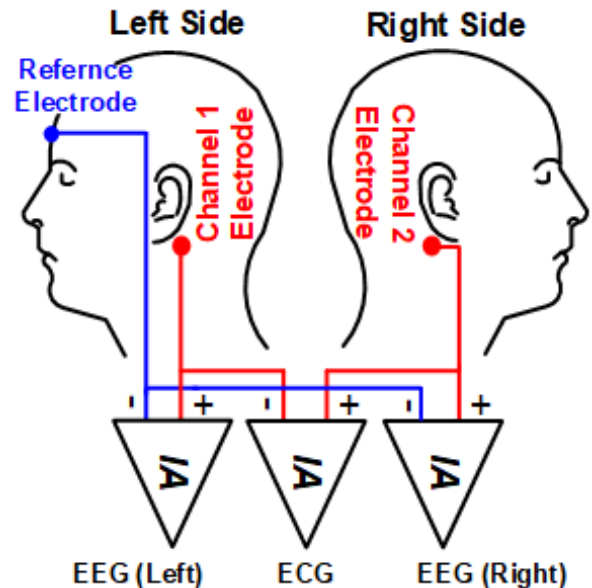


Fig. 1. Electrodes Placement

Differential of channel1 and channel2 with the reference electrode provides left side (TP9) and right side (TP10) EEG signals respectively. Taking differential of EEG channels (TP9 and TP10) approximates the ECG signal for a person. Three

AFE are designed for two EEG and one ECG channel. Three instrumentation amplifiers are used to measure the noise free EEG and ECG signals. The EEG and ECG recordings were verified using the alpha wave test and standard chest ECG measurement respectively. The DBE processor calculates the selected features including different band powers, R-R, and Q-S interval, which are forwarded to a shallow neural network (SNN) classifier for stress prediction.

II. DATA SET ACQUISITION

Stress experiments were performed in standard room conditions with adequate light and reduced noise conditions. 25 healthy subjects including 14 males and 11 females aged between 20-40 years participated in the stress test. The Mental Arithmetic Test (MAT) and Stroop Color-Word Test (SCWT) were used as the source of mild stress [12]. The EEG and ECG were recorded for the complete duration of the tests. Tests were performed in the following order: Mental Arithmetic Test (Stress1) – Rest (Rest1) - Stroop Color-Word Test (Stress2) - Rest (Rest2) as shown in Fig. 2.



Fig. 2. Stress Experiment Procedure

III. PROPOSED MENTAL STRESS CLASSIFICATION PROCESSOR

The proposed mental stress classification system consists of two major parts including an AFE amplifier for the EEG and ECG data acquisition and a DBE processor for the machine learning-based feature extraction and classification [13], [14].

Fig. 3 shows the overall block diagram of the proposed human mental stress classification system. It includes an AFE, analog to digital converter (ADC), micro-controller

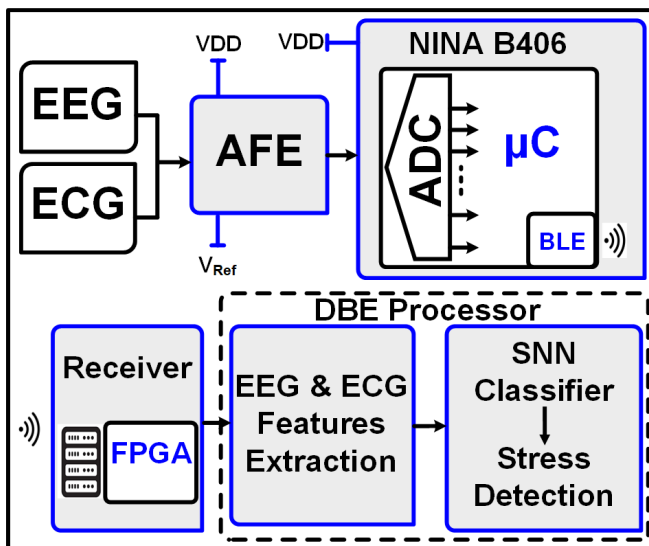


Fig. 3. Block Diagram of the Overall Mental Stress Classification System

(NINAB406), Bluetooth (BT) transceiver modules and a DBE processor. The EEG and ECG data is acquired through the AFE and forwarded to the ADC embedded in the Ublox NINA-B406 module. The NINA-B406 transmits the EEG and ECG data to the BT receiver integrated with the Xilinx Artix-7 FPGA. The EEG and ECG data is then utilized by the DBE processor for stress classification.

A. Analog Front End

The low power and low noise behind-the-ear wearable system is designed for the acquisition of EEG and ECG signals. The design of AFE is very critical to acquire the physiological signals with minimum noise and artifacts. The amplitude of EEG and ECG signals range in micro-volts(μV) and require amplification for stress prediction. A low-power AFE amplifier with a high signal-to-noise ratio (SNR) is designed for the amplification and filtration of the EEG and ECG signals. AFE consists of instrumentation amplifiers, band-pass, anti-aliasing filters and operational amplifiers [15].

Instrumentation amplifier amplifies the voltage difference of the input signals and rejects the common signal, hence noise is reduced significantly. Fig. 4 (a) shows the circuit diagram of the proposed AFE amplifier. A bandpass filter (BPF) is required to extract the required band of frequencies. The BPF is implemented using a high pass filter cascaded with a low pass filter. The high pass filter also acts as an anti-aliasing filter to remove the electrodes' DC offset. The low pass filter removes the noise and unnecessary frequency components.

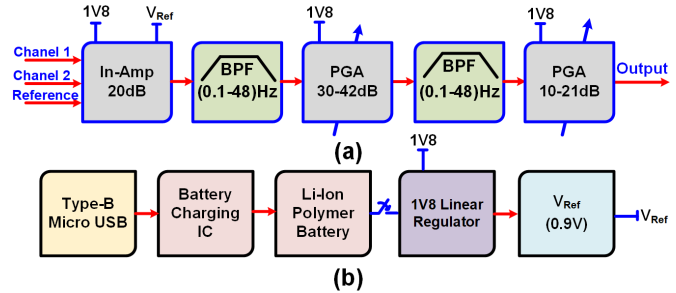


Fig. 4. (a) Circuit Diagram of AFE Amplifier (b) Power management for AFE amplifier

The OPA333 is used as an operational amplifier in the non-inverting configuration for the amplification of the filtered signal. The gain for this first stage OP-Amp is set to 42dB. A band-pass filter (0.1Hz to 48Hz) is also added after the first stage amplifier for acquiring noise-free signals. The second stage op-amp is used to further amplify the signals. (Channel 1 & Reference Channel), (Channel 2 & Reference Channel) and (Channel 1 & Channel 2) pairs acquire/estimate the data for Left Side EEG, Right Side EEG and ECG respectively.

Fig. 4 (b) shows the power management for the AFE. A 3.7V 300 mAh Li-Ion polymer battery along with a secondary recharging integrated circuitry and a standard micro-Type-B USB is used to power the system. The supply voltage is reduced to 1V8 in order to reduce the power consumption

of the system. A 1V8 linear regulator (STLQ020C18RV) with ultra-low quiescent current ($0.3\mu A$) is used to provide a constant 1V8. A reference voltage of 0V9 is provided using a voltage divider followed by a low impedance op-amp buffer.

Fig. 5 shows the gain bandwidth (GBW) transfer function of EEG and ECG AFE amplifiers. In the case of EEG Signals, a gain of 82dB was obtained with low and high pass frequencies of 0.1Hz and 48Hz respectively. Similarly, a gain of 66dB was used for ECG signals. The noise of the designed amplifiers is also measured. A total noise of $0.1\mu V_{rms}$ was obtained in the designed frequency band (0.1Hz-48Hz).

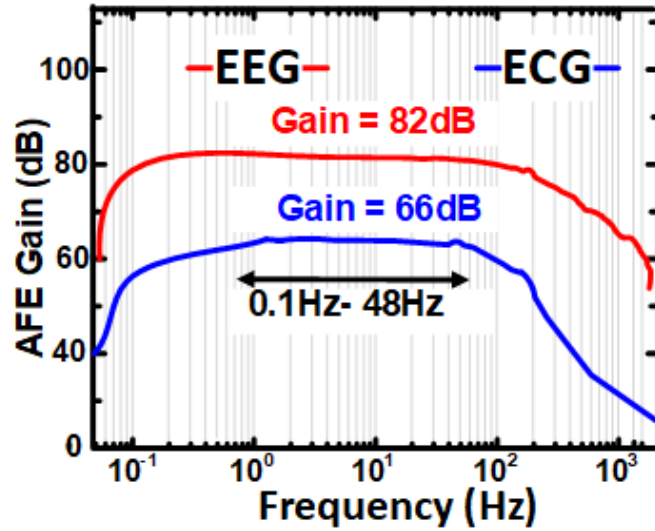


Fig. 5. Gain Bandwidth (GBW) Transfer Function Plots of EEG and ECG AFE Amplifiers

The amplified EEG and ECG signals from the AFE are forwarded to a 12-bit successive approximation analog to a digital converter (SAR ADC). The digitized EEG and ECG signals are then transmitted wirelessly using the Ublox NINA-B406 module. NINA-B406 is selected due to its low power consumption (1.7V & 14mA) and stand-alone properties. It is built from nRF52833 SOC that comprises a powerful Arm Cortex M4 along with RF Core.

The EEG and ECG signals acquired using the proposed AFE are received by the Spartan 3-E FPGA using a BT receiver. The signals are then forwarded to the DBE processor for the mental stress classification.

B. DBE Processor

The DBE processor contains a feature extraction unit (FEU) and a machine learning classifier to predict mental stress. The FEU calculates the selected features using the EEG and ECG signals. The EEG signals are used to calculate the band powers in the frequency bands including 0-4 Hz, 4-8Hz, 8-13 Hz, and 13-30 Hz [16]. The ECG-based features include the band powers in the frequency bands including 0.01-0.15Hz, 0.15-0.4 Hz, RR, and QS interval [17].

Fig. 6 highlights the selected features calculated by the FEU. The FIR filters of the 100th orders were used to calculate

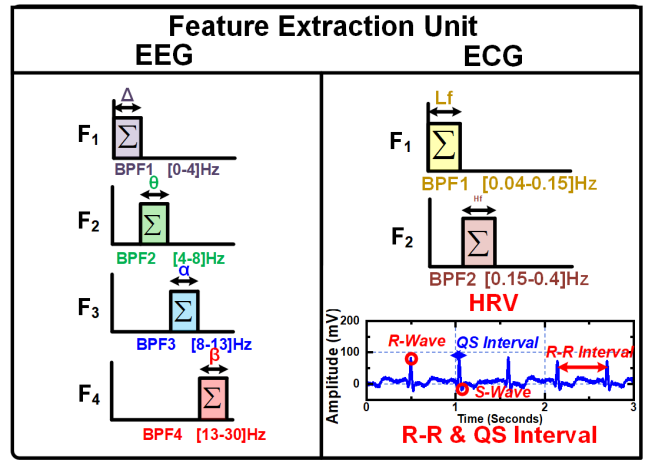


Fig. 6. Feature Extraction Unit

the band energies for the EEG signals. Pan and Tomkins's algorithms were used to calculate the RR and QS interval [17]. The ECG signals were re-sampled to obtain uniform space intervals. The SNN classifier provides a classification accuracy of 92.7%.

Fig. 7 shows the architecture of the proposed SNN classifier for the stress prediction. The feature vectors (FV0-FV11) are forwarded to the SNN classification unit. The SNN classification unit selects a subset of patient-specific 8 features (F0-F7) using a feature selector and a feature mask. The subset of 8 features is forwarded to the SNN classification unit for stress prediction. The SNN classification unit is implemented in a fully pipelined and customized manner. Two multiplexers are used to select the weights and biases using control units (control_1 and control_2). The SNN classifier unit multiplies and accumulates the input features and then applies the sigmoid activation function to predict a patient as stressed or non-stressed.

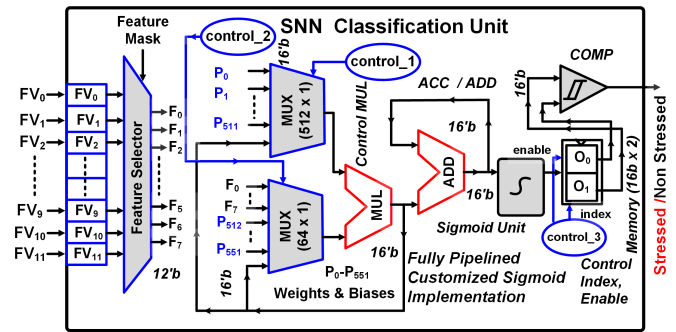


Fig. 7. SNN Classifier Unit

IV. MEASUREMENTS AND RESULTS

A. EEG Verification Test

Alpha wave test was performed for EEG signals verification. Fig. 8 shows the raw EEG during the closed and opened eye states. A significant increase in the amplitude of EEG can be

observed during the closed eyes state due to the presence of the alpha wave. Power spectrum analysis was also performed and it was observed that the alpha wave frequency component (8Hz-13Hz) increases during the closed eye state.

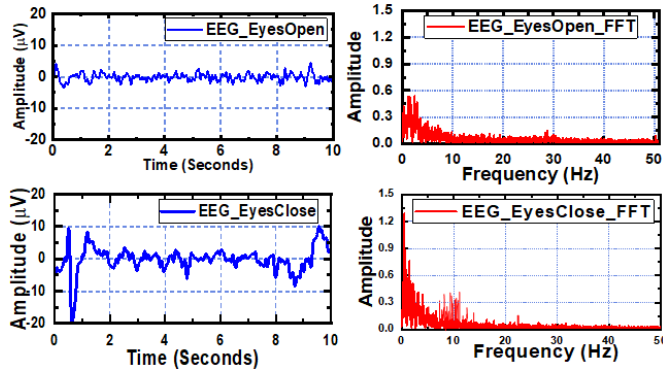


Fig. 8. Alpha Wave Test Results

B. ECG Verification Test

Standard chest ECG and estimated ECG using EEG signals were measured and compared simultaneously for 60 seconds. Fig. 9 (a) shows the standard ECG measured from the chest (red) and the estimated ECG measured using EEG signals (blue). Fig. 9 (b) shows the fast Fourier transforms (FFT) for both ECG waveforms. It can be observed that the estimated ECG signal along with its FFT is similar to the standard ECG signal.

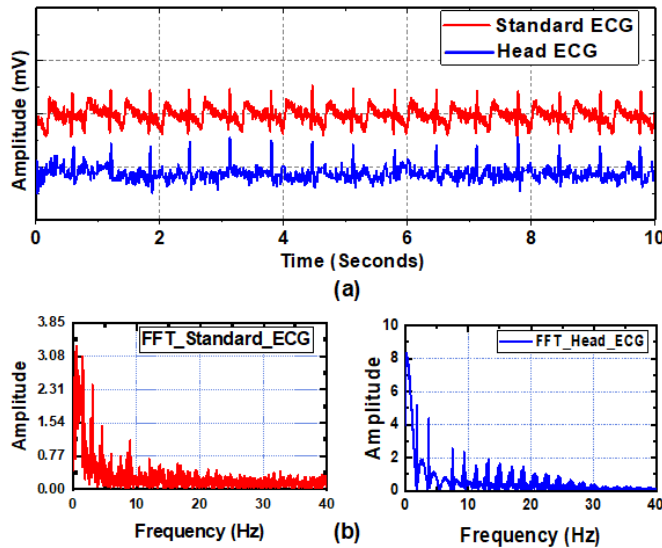


Fig. 9. ECG Test Results

V. RESULTS AND DISCUSSION

There is very limited previous research available for mental stress detection using multiple physiological signals (EEG+ECG) simultaneously using a single device [18]. Majority of the previous works either lack multiple physiological

signals or utilize them independently [19], [20]. Similarly, majority of the previous works lack an online and real-time stress classification digital back-end processor. In this work, we have presented a novel wireless wearable device that measures EEG and ECG simultaneously using only three electrodes along with a hardware based mental stress classification processor [21].

Table I summarizes the comparison of this work with the state-of-the-art works. This work utilizes both EEG and ECG signals for the mental stress classification and provides AFE and a DBE processor. The AFE records the EEG and ECG data with minimum noise and the DBE classifies the human stress with maximum classification accuracy. This is the 1st work to the best of our knowledge to target a stand-alone system for the mental stress classification. It utilizes a minimum number of channels (2-EEG, 1-ECG), provides the best classification accuracy (92.7%), has minimum noise level ($0.1\mu V_{rms}$), minimum current (8mA), minimum power (14.4mW) and provides both AFE and DBE.

TABLE I
COMPARISON WITH THE STATE-OF-THE-ART-WORKS

Parameters	Sensors'19 [18]	EMBC'19 [19]	EMBC'19 [20]	CICC'20 [10]	ICSIMA'21 [16]	This Work
Physiological Signals	EEG+ECG	EEG+GSR+PPG	ECG+PPG	EEG	EEG	EEG+ECG
Channels Count	2+1	4+1+1	4+1	2	7	2+1
Classifier	SVM	MLP	RF	DNN	SVM	SNN
Accuracy	87.5%	75%	84.13%	85%	90%	92.7%
Noise Level	$0.12\mu V_{rms}$	----	----	$0.55\mu V_{rms}$	$0.12\mu V_{rms}$	$0.10\mu V_{rms}$
Current (mA)	9.6mA	----	----	9.6mA	---	8mA
Power (mW)	17.28mW	----	----	----	----	14.4mW
AFE+DBE	No	No	No	No	----	Yes

VI. CONCLUSION

This paper presents a novel wireless wearable device that measures EEG and ECG simultaneously using only three non-invasive electrodes. The developed device has a small form factor, is lightweight and easily behind-the-ear wearable. It is ultra-low power and provides excellent noise performance ($0.1\mu V_{rms}$) for a bandwidth of 0.1Hz - 48Hz. The measurements obtained from the device are also verified. Alpha wave test is performed to verify the EEG signals while obtained ECG signals are compared with standard chest ECG. Various EEG and ECG features are used for the mental stress classification with an excellent classification accuracy (92.7%).

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